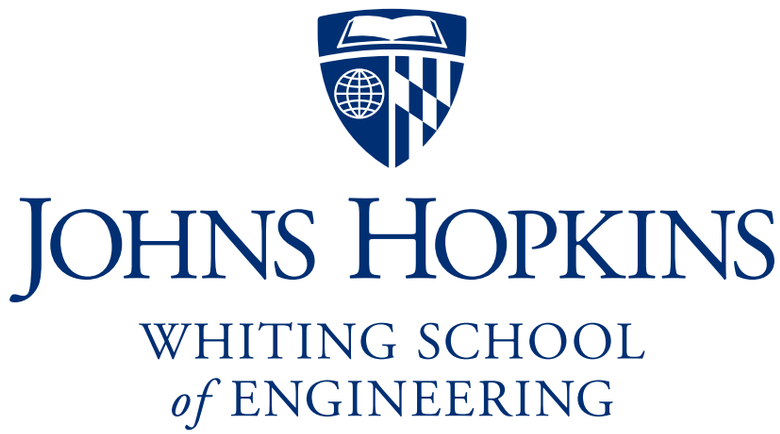
Sketching k-mers

Ben Langmead
JHU Computer Science
langmea@cs.jhu.edu, langmead-lab.org

Scalable Genomics and Pangenomics course, 2026



Sketching

How do we sift and summarize huge datasets so we can answer similarity questions later?

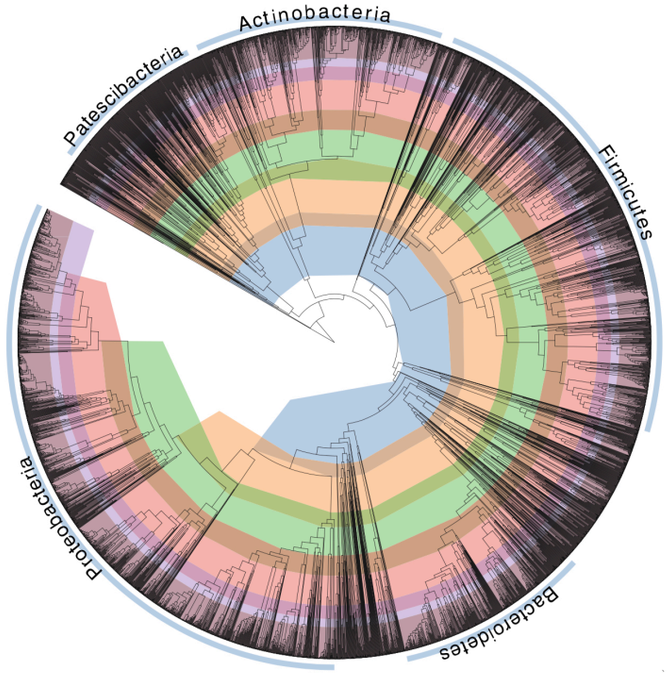


Image: doi:10.1038/nbt.4229

Genomes

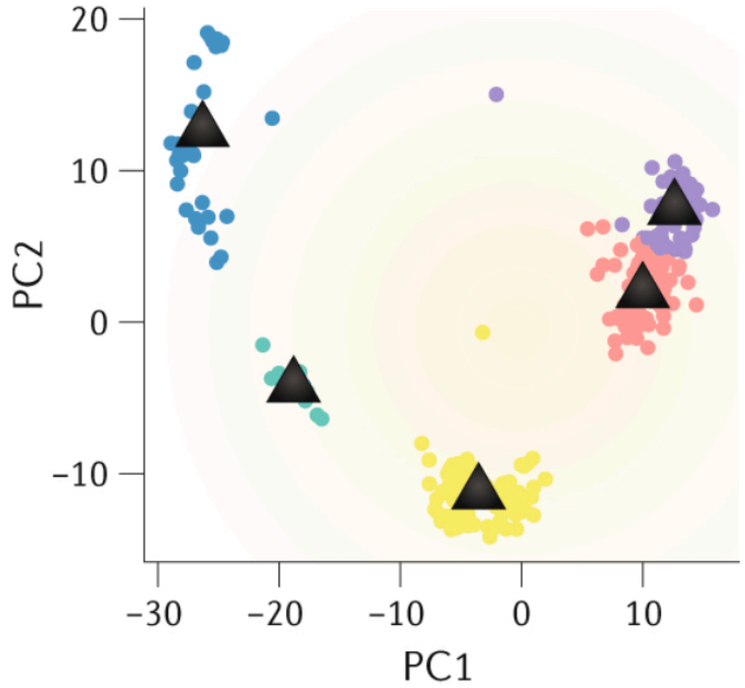


Image: doi:10.1038/s41576-018-0088-9

Sequencing data

k-mers

AGGCCACAGTGTATTATGACTGAAAAAAAAAAAAGATGTTCAGATTCACAGCCA



AGGCCACAGT TATTATGACT AAAAAAAAAA GATGTTCAGA
 GGCCACAGTG ATTATGACTG AAAAAAAAAA ATGTTCAGAT
 GCCACAGTGT TTATGACTGA AAAAAAAAAG TGTTCAGATT
 CCACAGTGTA TATGACTGAA AAAAAAAAGA GTTCAGATTC
 CACAGTGTAT ATGACTGAAA AAAAAAAGAT TTCAGATTCA
 ACAGTGTATT TGACTGAAAA AAAAAAGATG TCAGATTCAC
 CAGTGTATTA GACTGAAAAA AAAAAGATGT CAGATTCACA
 AGTGTATTAT ACTGAAAAAA AAAAGATGTT AGATTCACAG
 GTGTATTATG CTGAAAAAAA AAAGATGTTC GATTCACAGC
 TGTATTATGA TGAAAAAAAA AAGATGTTCA ATTCACAGCC
 GTATTATGAC GAAAAAAAAA AGATGTTCAG TTCACAGCCA

k-mers

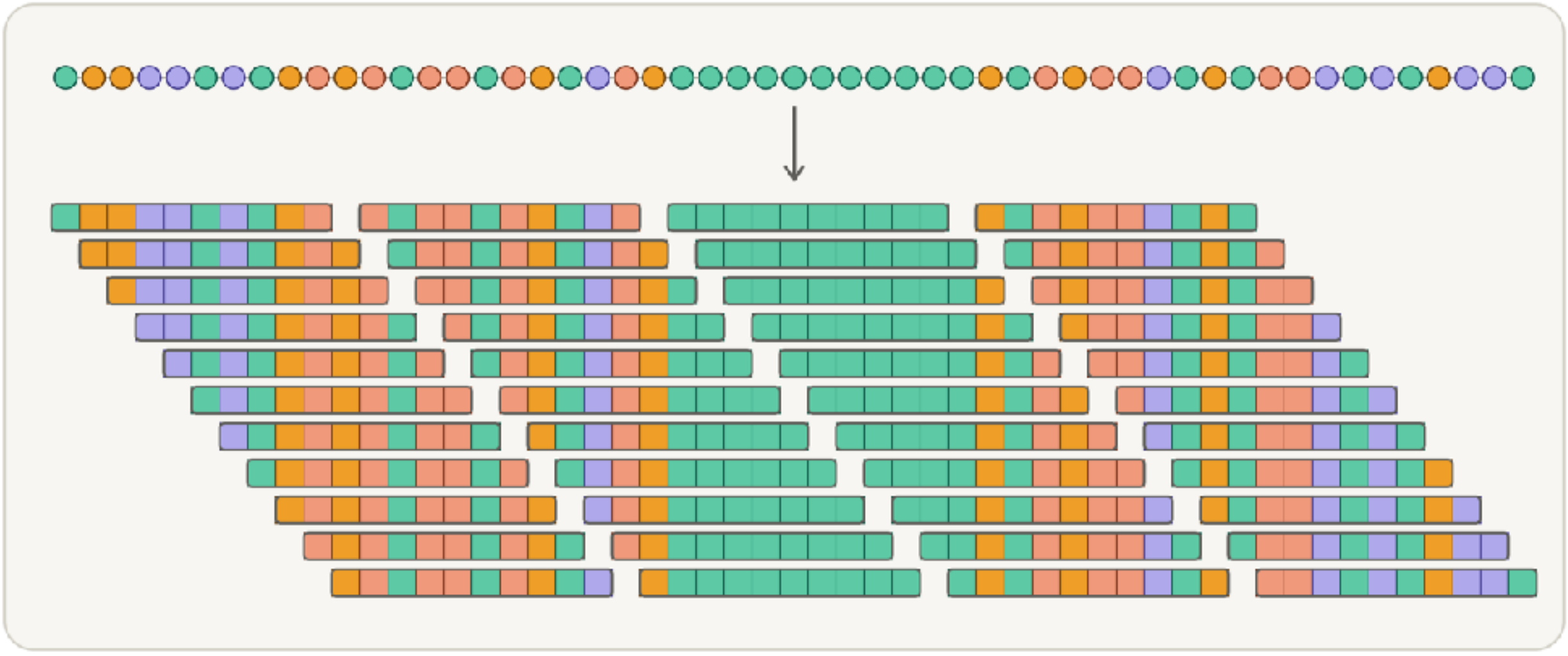
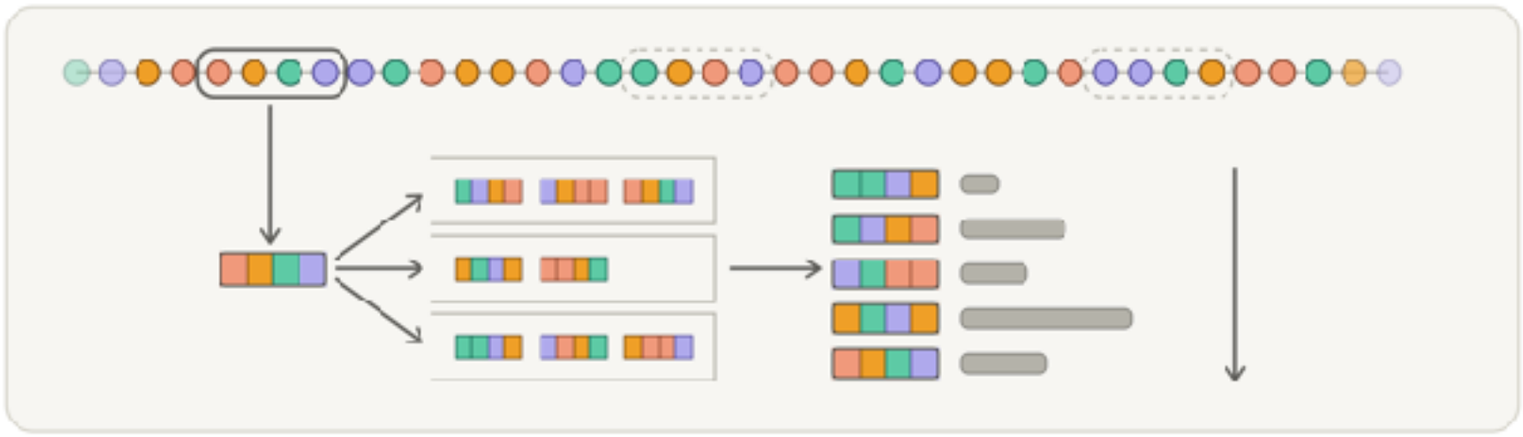
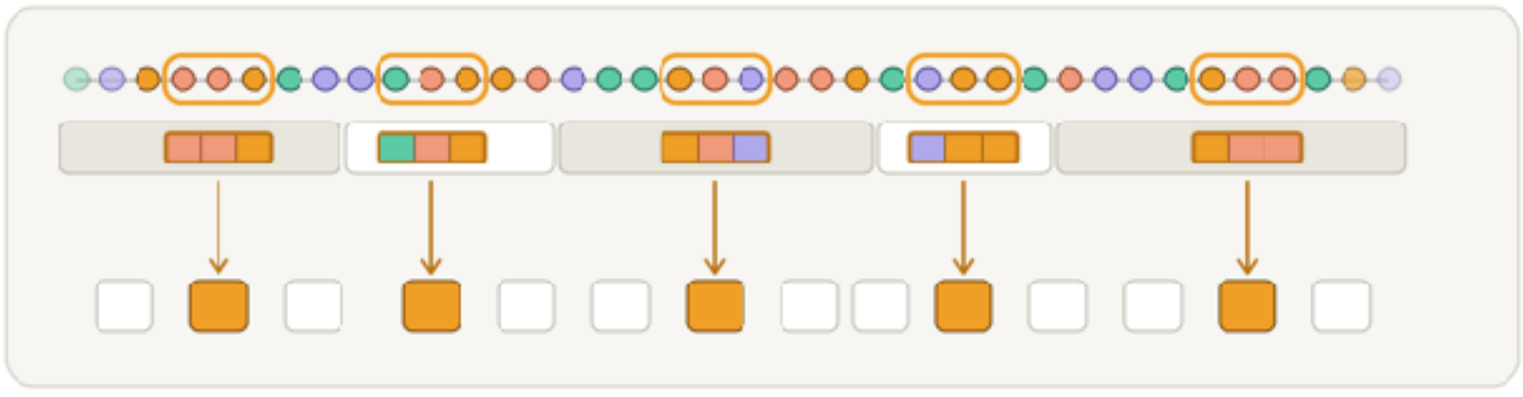


Image by Ben & Claude Sonnet 5.5

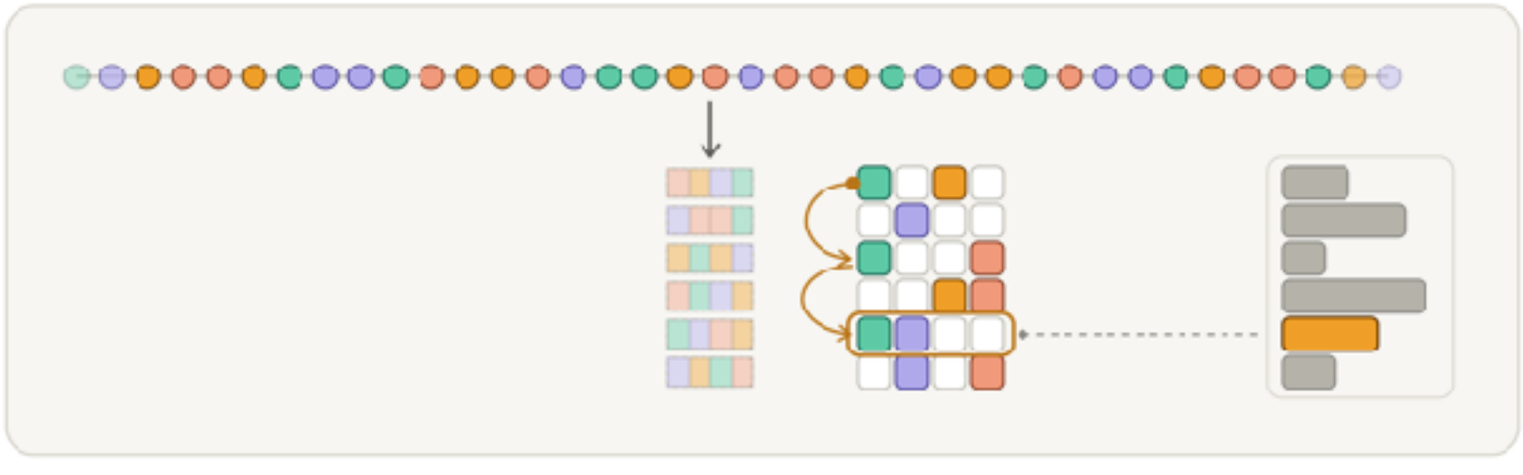
Sets of k-mers



KMC (sorting)



SSHash
(grouping with minimizers)



SBWT
(fancy sorting)

Image by Ben & Claude Sonnet 5.5

Sets of k-mers

A **computational expedient**

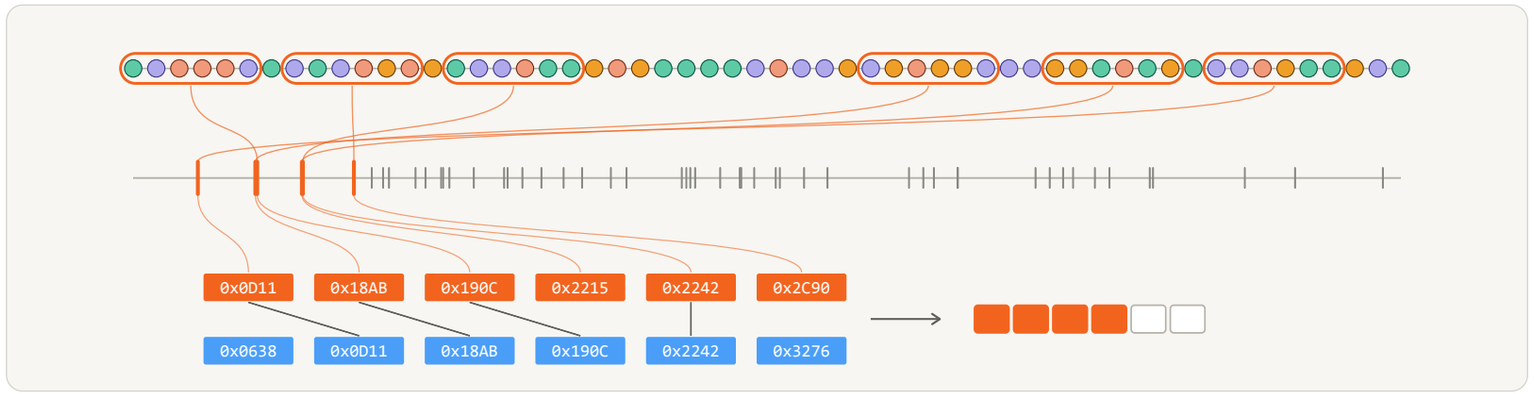
not “the truth”

Sketches of sets of k-mers

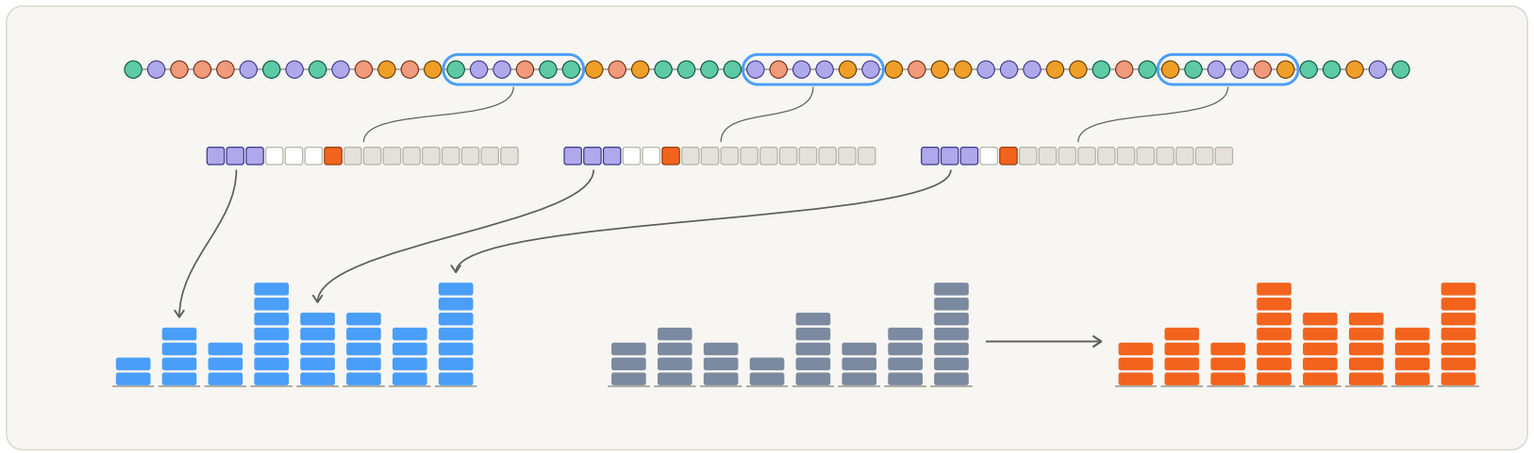
A **super-expedient**

still further from “the truth”

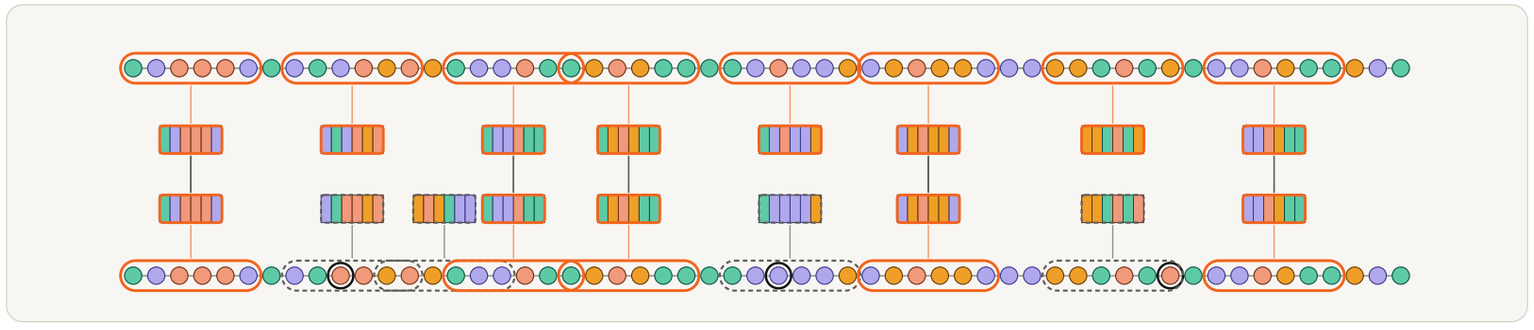
Sketches of sets of k-mers



Mash
(MinHash)



Dashing
(HyperLogLog)

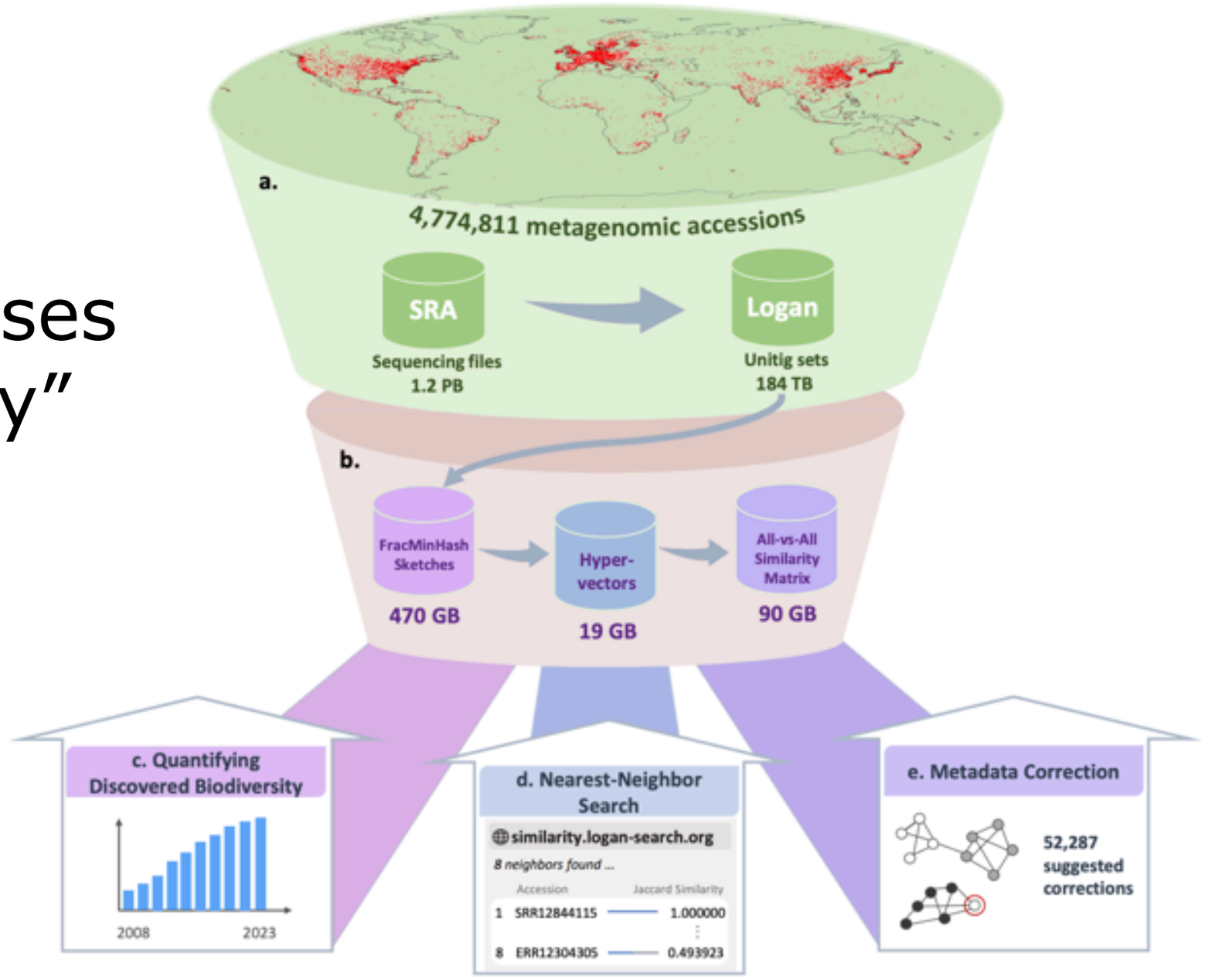


Minimap
(minimizers)

Image by Ben & Claude Sonnet 5.5

Sketching

Lifts k-mer based analyses to “planetary” scale



Sketching the SRA: experiment-wide similarity search across 4.8 million metagenomes

Roland Faure, Adrita Hossain Nakshi, Md. Hasin Abrar, Rayan Chikhi, Teo Lemane, Mohsen Taheri, Stephanie Won, Haonan Wu, David Koslicki, Paul Medvedev

bioRxiv 2026.10.07.757297

Sketching





Cardinality

Cardinality & coincidence

$$|A|$$

$$|B|$$

$$|A \cup B|$$

$$|A \cap B|$$

Set similarity

$$C = \frac{|A \cap B|}{|A|}$$

$$J = \frac{|A \cap B|}{|A \cup B|}$$

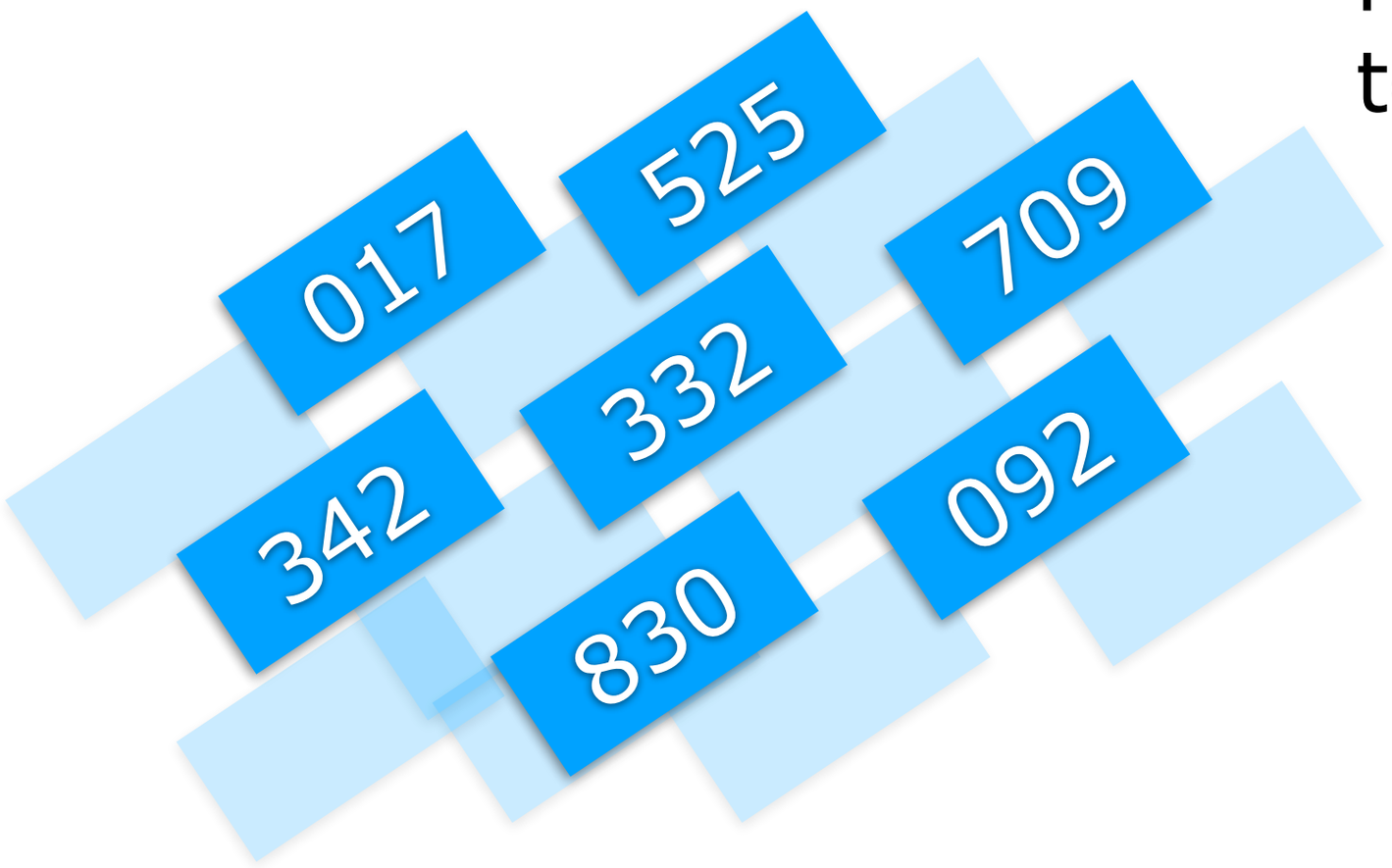
Biological relatedness

```
AGGCCACAGTGTATTATGACTG
|||||||||||  |||||||||
AGGCCACAGTGAGTTATGACTG

AAAAAAAAAAAGATGT-AAGTA
|||||||||||||||| |||||
AAAAAAAAAAAGATGTAAAGTA

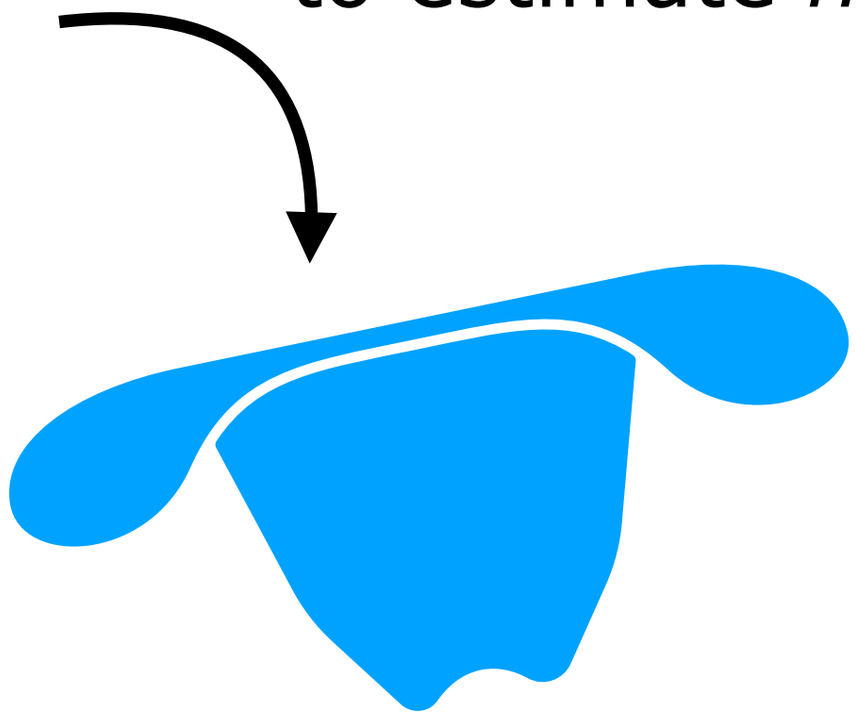
GAGG--TCAGATTCACAGCCAC
||||  ||||||||||||||||
GAGGGGTCAGATTCACAGCCAC
```

Hat problem



I take cards labeled 1--1,000 and choose a random subset of size *n* to hide in my hat

You would like to estimate *n*

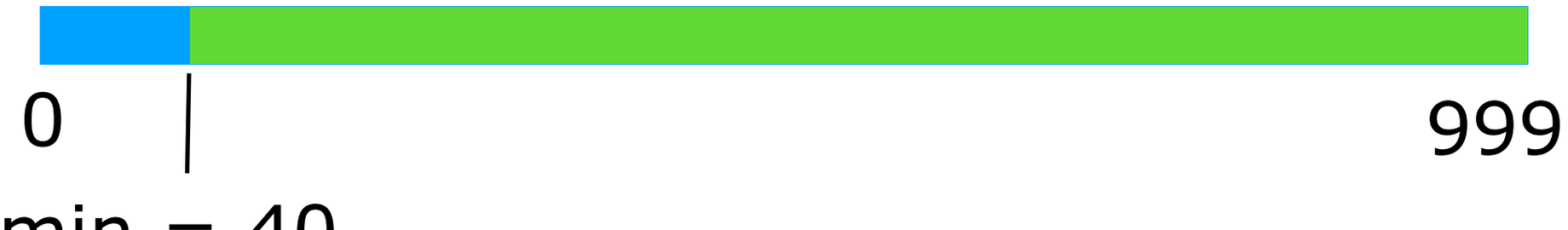


You may see **one representative** from the cards in the hat; which to pick?

Hat problem

What if **minimum** was 500? ...10? ... 4?

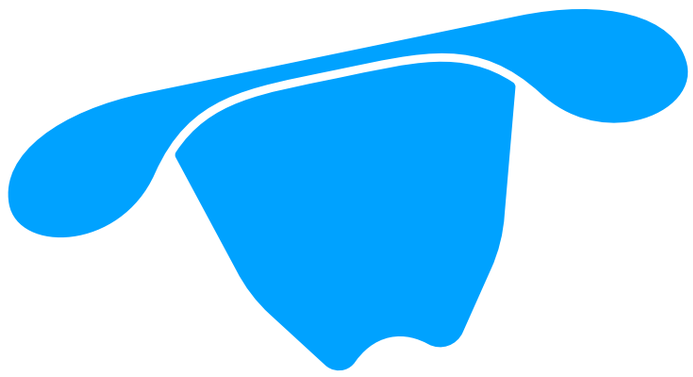
Estimate should grow as minimum shrinks



min = 40

$$40 \approx 1000/(n + 1)$$

$$n \approx 24$$



Easy to compute, fits in 10 bits

Hash functions

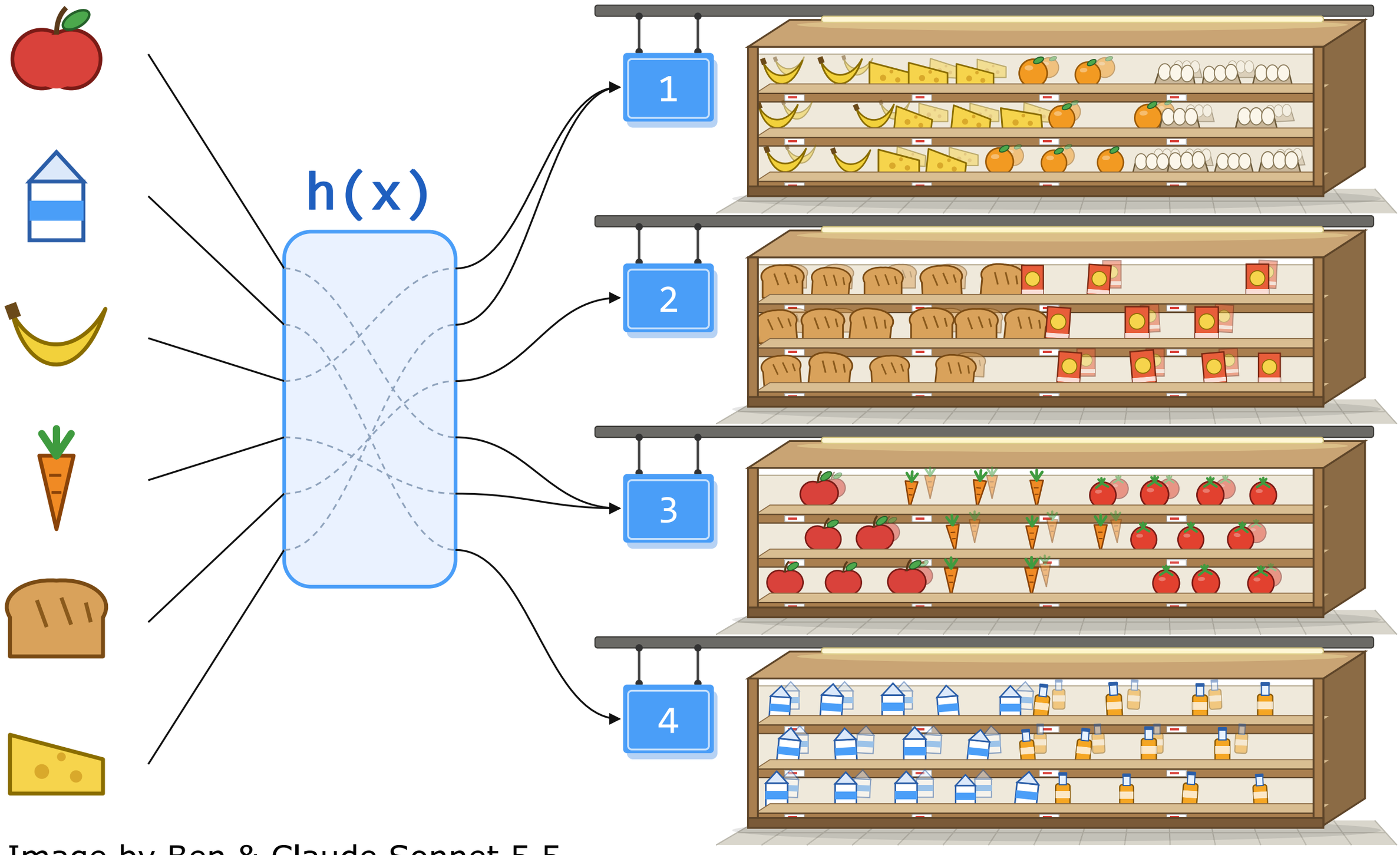


Image by Ben & Claude Sonnet 5.5

Hash functions

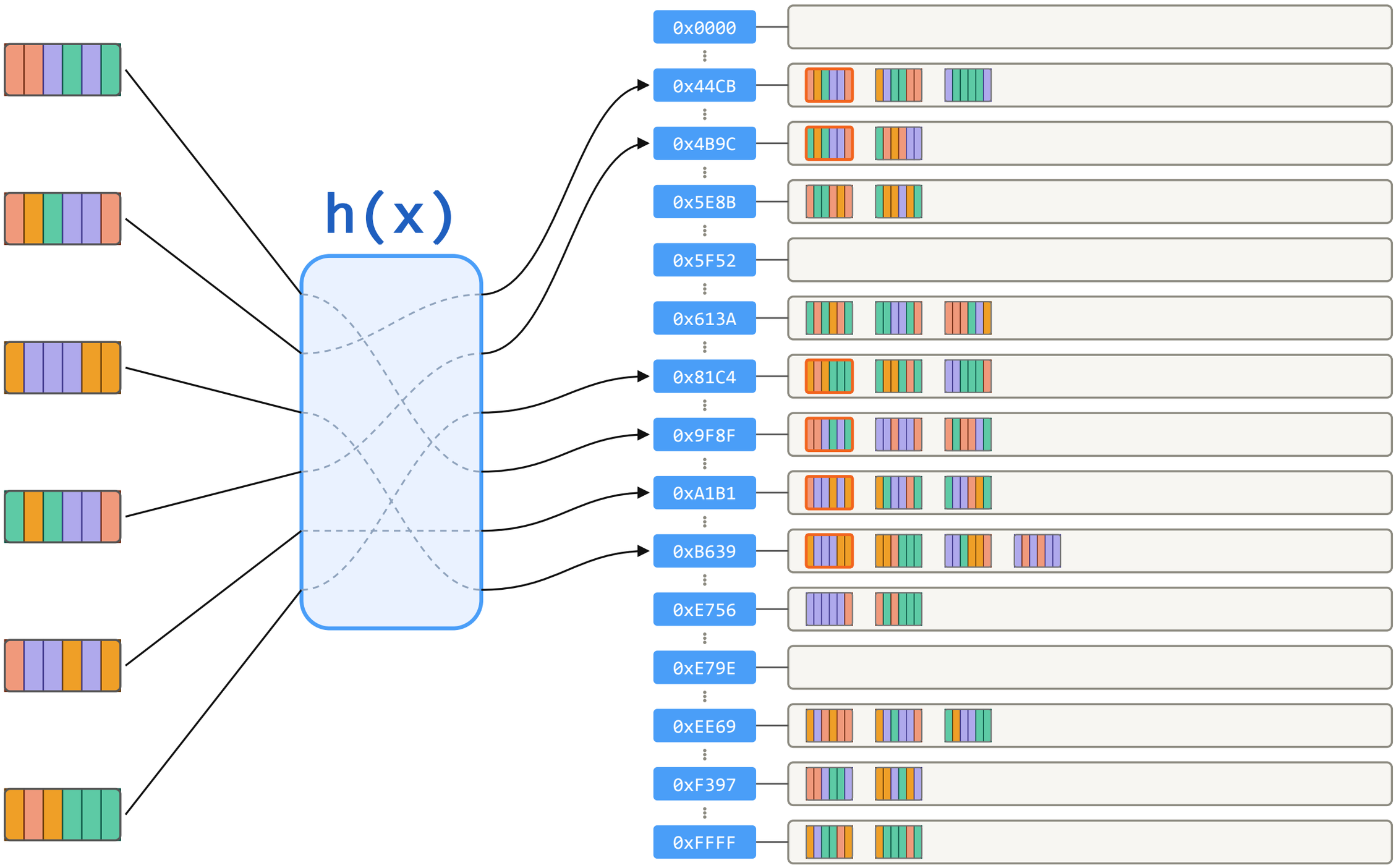
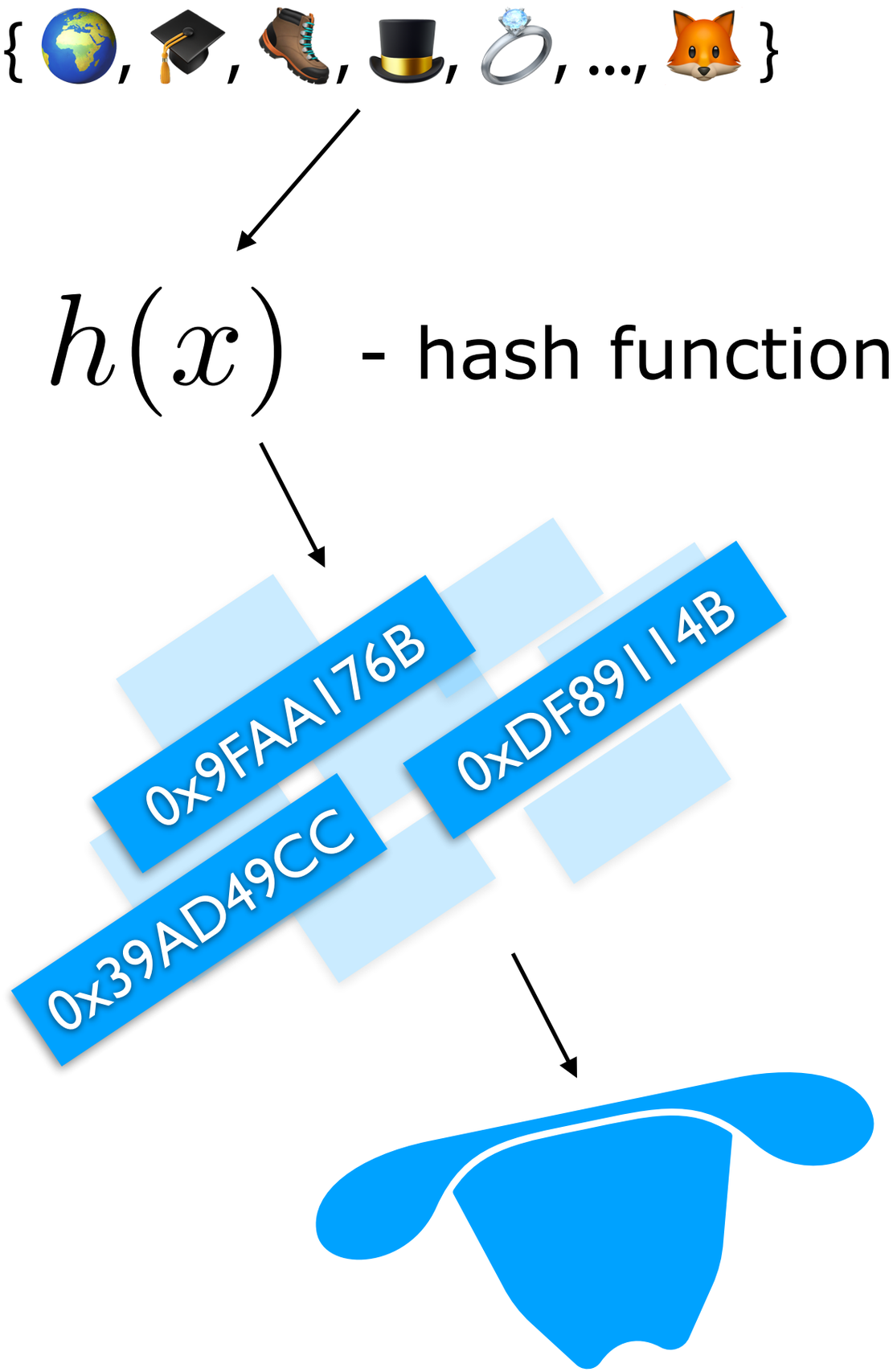


Image by Ben & Claude Sonnet 5.5

Hat problem

Hat analogy *seems* contrived...

...but matches the situation where we *hash* items up front



Two-hat problem

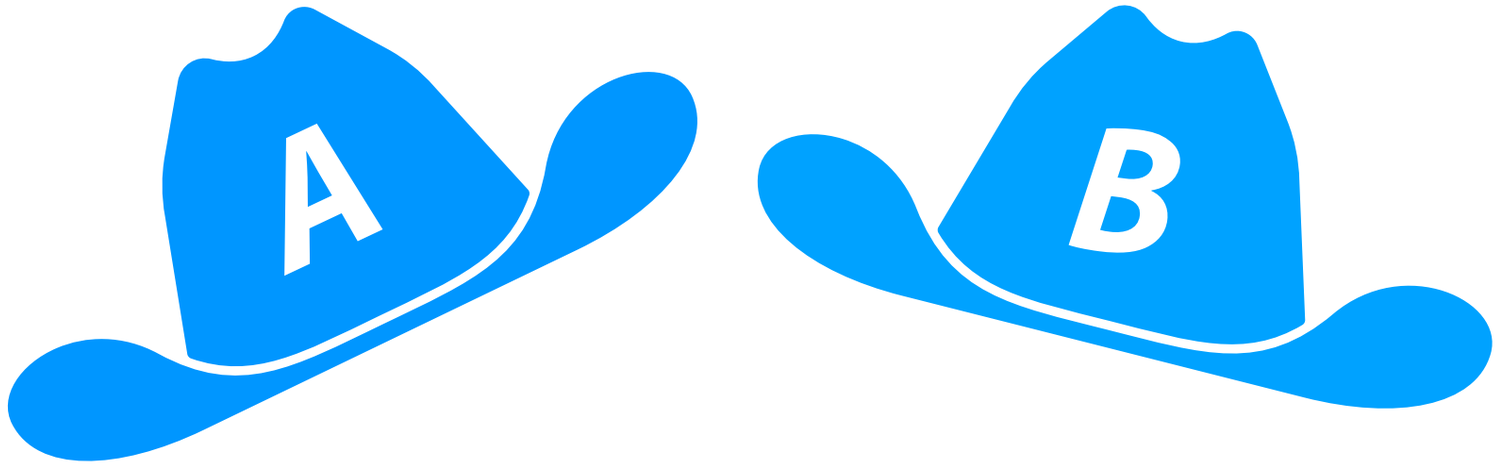
Cardinalities

$|A|$

$|B|$

$|A \cup B|$

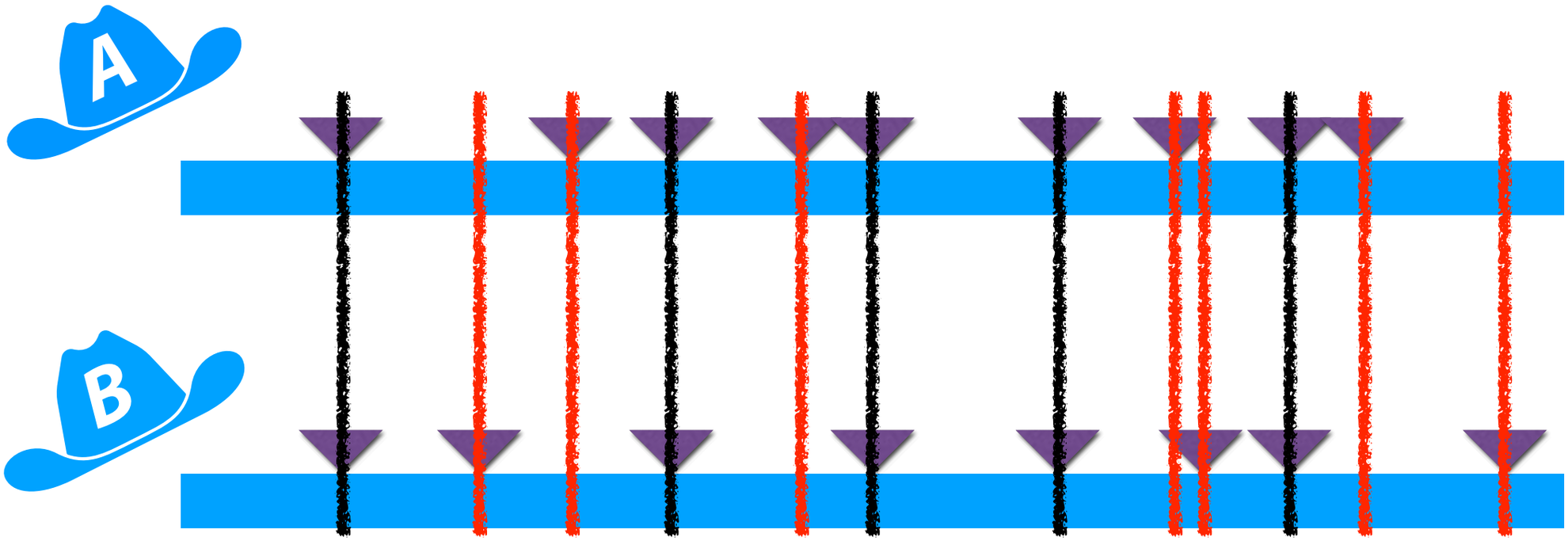
$|A \cap B|$



Can we estimate cardinality of unions? Intersections?

About *coincidences*

Two-hat problem

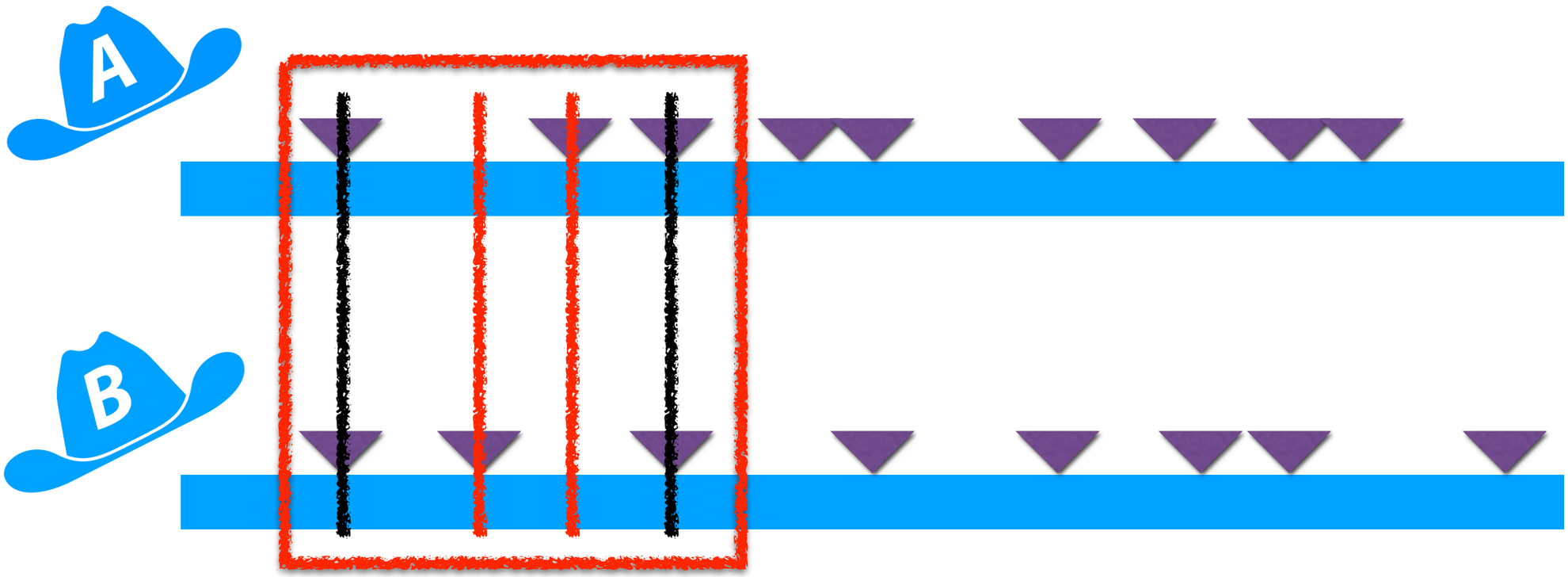


Space of coincidences is large

Need multiple representatives per set

Image inspired by: Ondov B, Starrett G, Sappington A, Kostic A, Koren S, Buck CB, Phillippy AM. **Mash Screen: high-throughput sequence containment estimation for genome discovery**. *Genome Biol* 20, 232 (2019)

Bottom k



Instead of minimum only, consider "bottom 3"

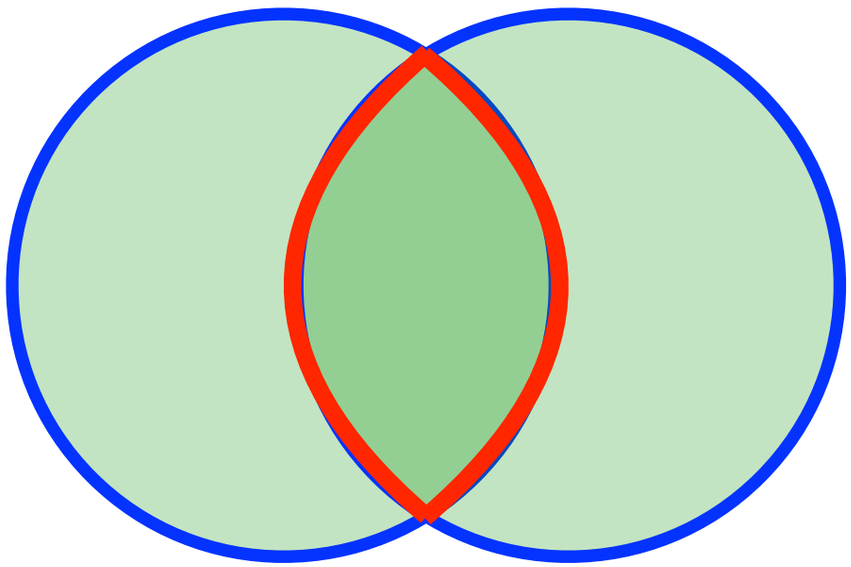
"Bottom-k sketch" can estimate cardinalities of unions and intersections

Larger k averages more, improving estimate

Jaccard coefficient

To measure similarity of A & B, we're interested in the size of $A \cap B$ i.e. $|A \cap B|$

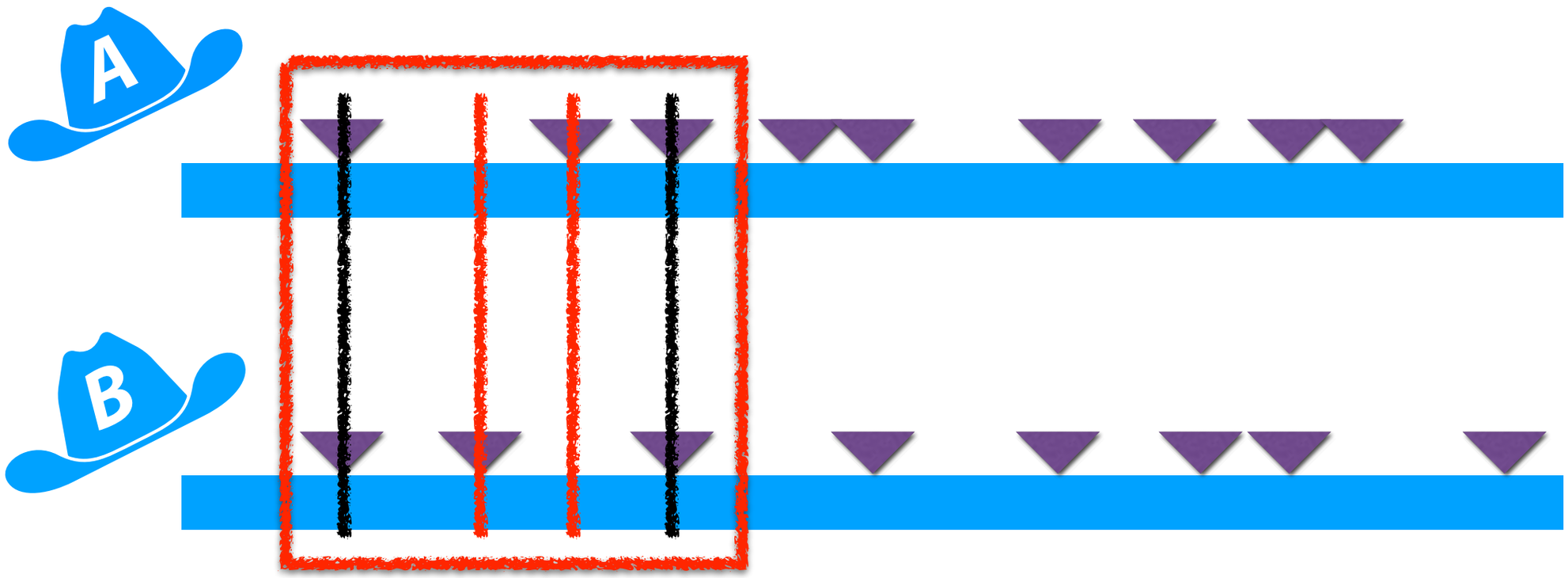
This is best understood relative to the sizes of the sets themselves, so we divide by $|A \cup B|$

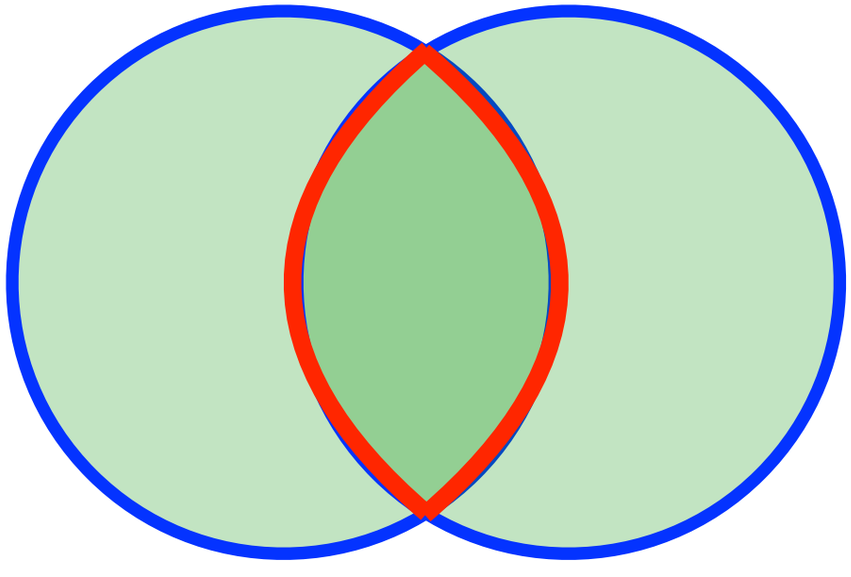


$$\frac{|A \cap B|}{|A \cup B|} = J$$

J is the *Jaccard coefficient*

Jaccard coefficient





$$\frac{|A \cap B|}{|A \cup B|} = J$$

J is the *Jaccard coefficient*

Mash

Mash sketch =
Bottom-k
MinHash sketch

Default k = 1000

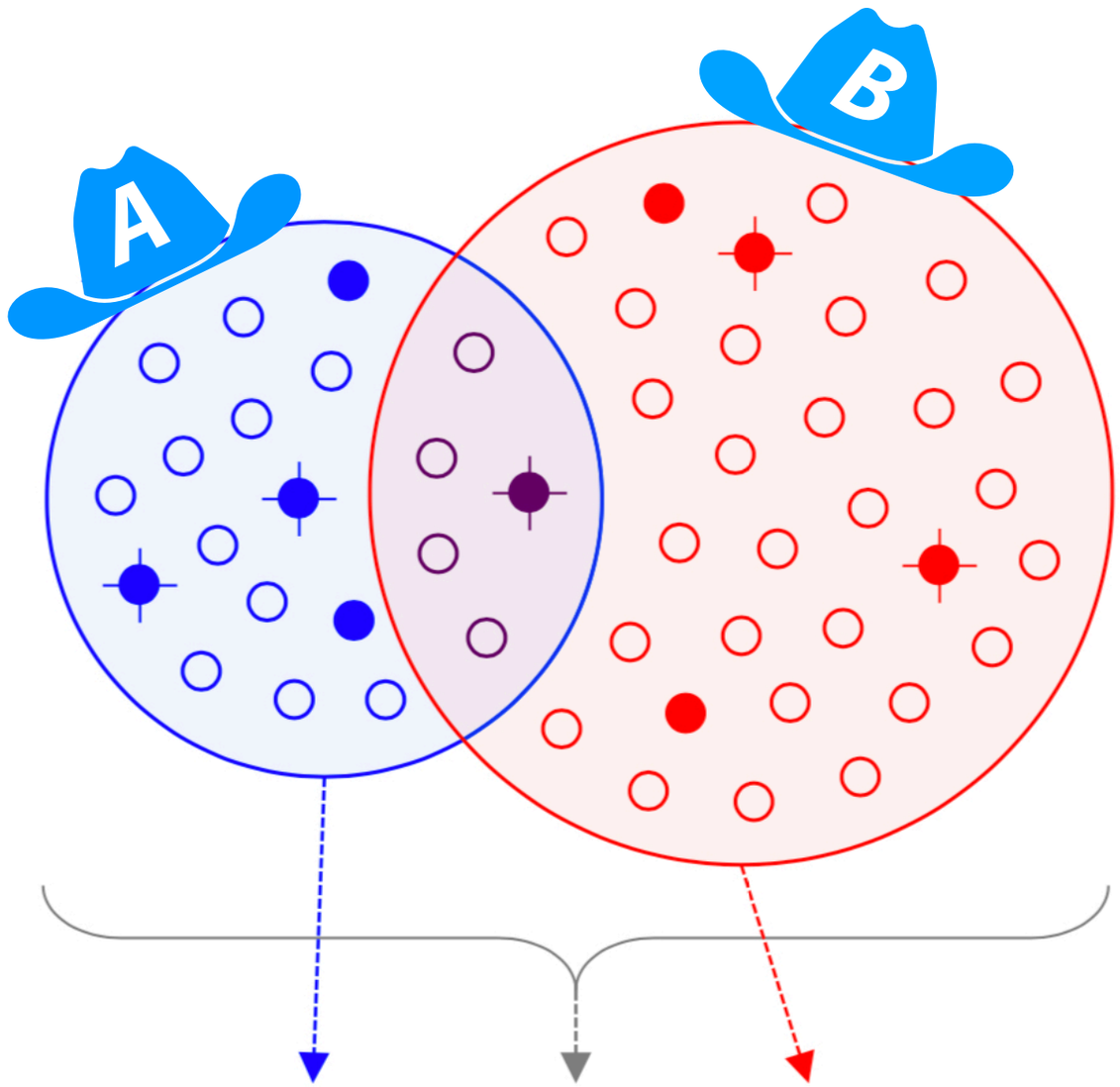


Image: ref 1

1. Ondov BD, Treangen TJ, Melsted P, Mallonee AB, Bergman NH, Koren S, Phillippy AM. **Mash: fast genome and metagenome distance estimation using MinHash**. *Genome Biol*. 2016 Jun 20;17(1):132.

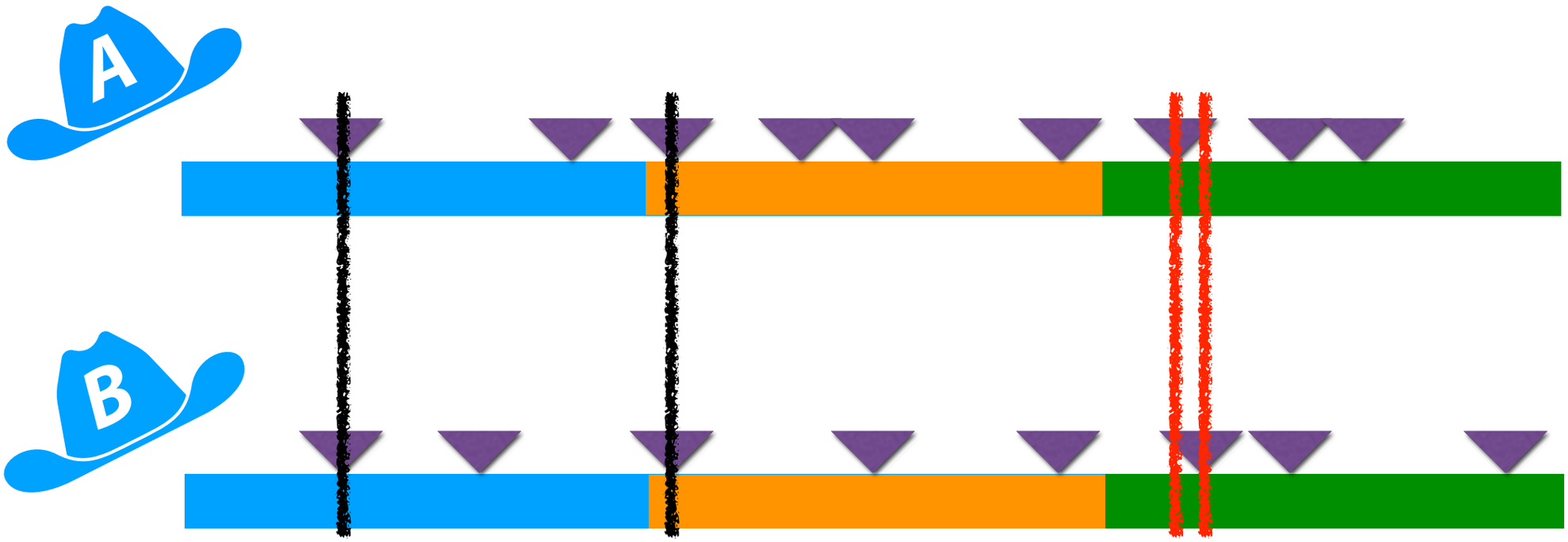
2. Broder AZ. **On the resemblance and containment of documents**. Compression and Complexity of Sequences 1997 - Proceedings 1998:21–29.

Origins of bottom-k MinHash



AltaVista in 1999, as seen in Netscape web browser

k-partition



Instead of bottom-3, consider the individual minimum in each of 3 “partitions”

Extremely similar to bottom-k

Mash

$S(A)$	$S(A \cup B)$	$S(B)$
42	42	66
64	64	82
82	66	87
128	**82**	104
139	87	127

Image: ref 1

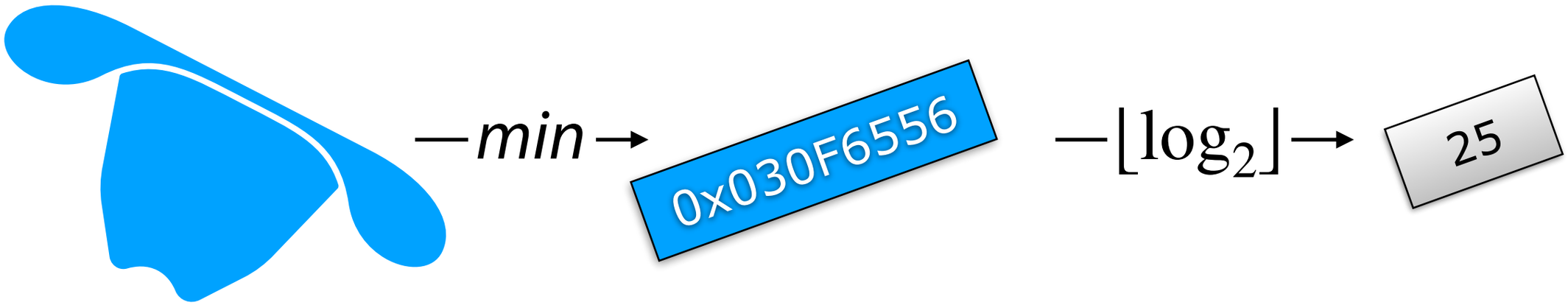
Representative fits in $\lceil \log_2 U \rceil$ bits

U is often 32 or 64

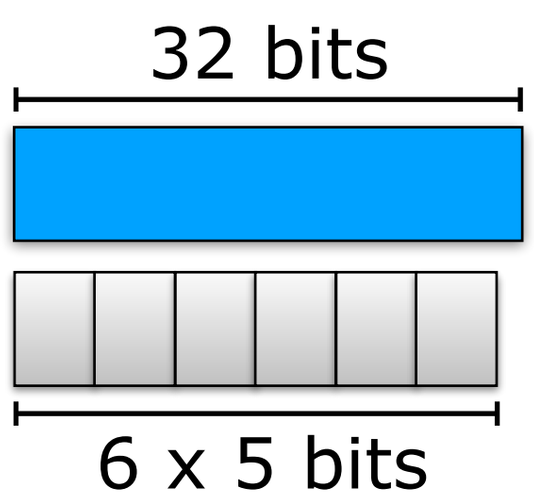
1. Ondov BD, Treangen TJ, Melsted P, Mallonee AB, Bergman NH, Koren S, Phillippy AM. **Mash: fast genome and metagenome distance estimation using MinHash**. *Genome Biol*. 2016 Jun 20;17(1):132.

From Log to LogLog

Instead of *minimum*, say we use *log-minimum*



Pro: Representatives take $\log \log U$ rather than $\log U$ bits



Con: Estimate is of $\lfloor \log_2 n \rfloor$; can re-exponentiate later, but with added variance & bias

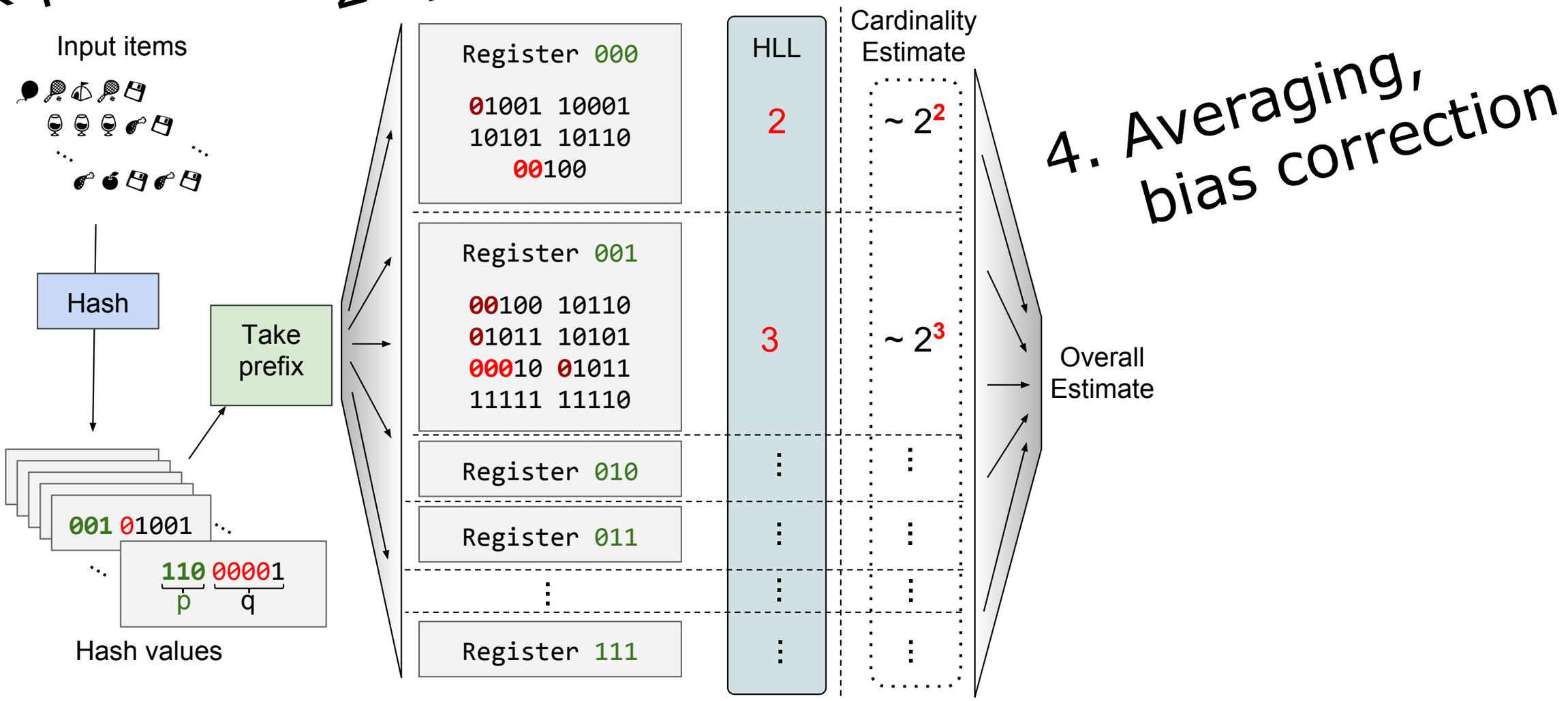
HyperLogLog

1. k-partition

2. $\lfloor \log_2 n \rfloor$

3. Re-exponentiation

4. Averaging, bias correction



Baker, DN., Langmead B. **Dashing: fast and accurate genomic distances with HyperLogLog**. *Genome biology* 20.1 (2019): 265.

Baker DN., Langmead B. **Genomic sketching with multiplicities and locality-sensitive hashing using Dashing 2**. *Genome Research* 33.7 (2023): 1218-1227.

Sketching

Fixed, small space budget

Output is a “sketch” that packs the estimators in as tight as possible

Sketches tell us about cardinalities and coincidences…

…which in turn tell us about sequence similarity

Mash, Dashing, Bindash, …

Digesting

Another kind of “sketching”… or “digesting”

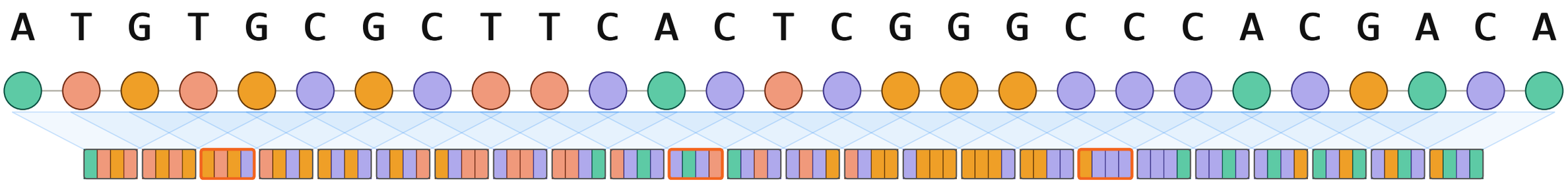


Image by Ben & Claude Sonnet 5.5

Another kind of “sketching”… or “digesting”

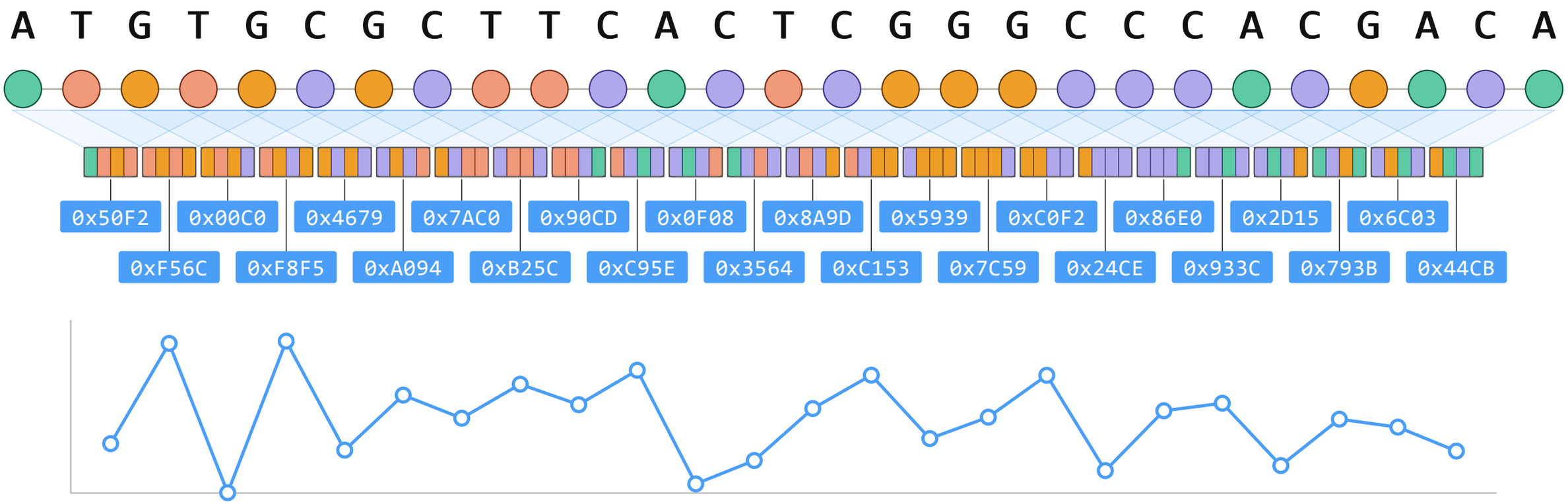
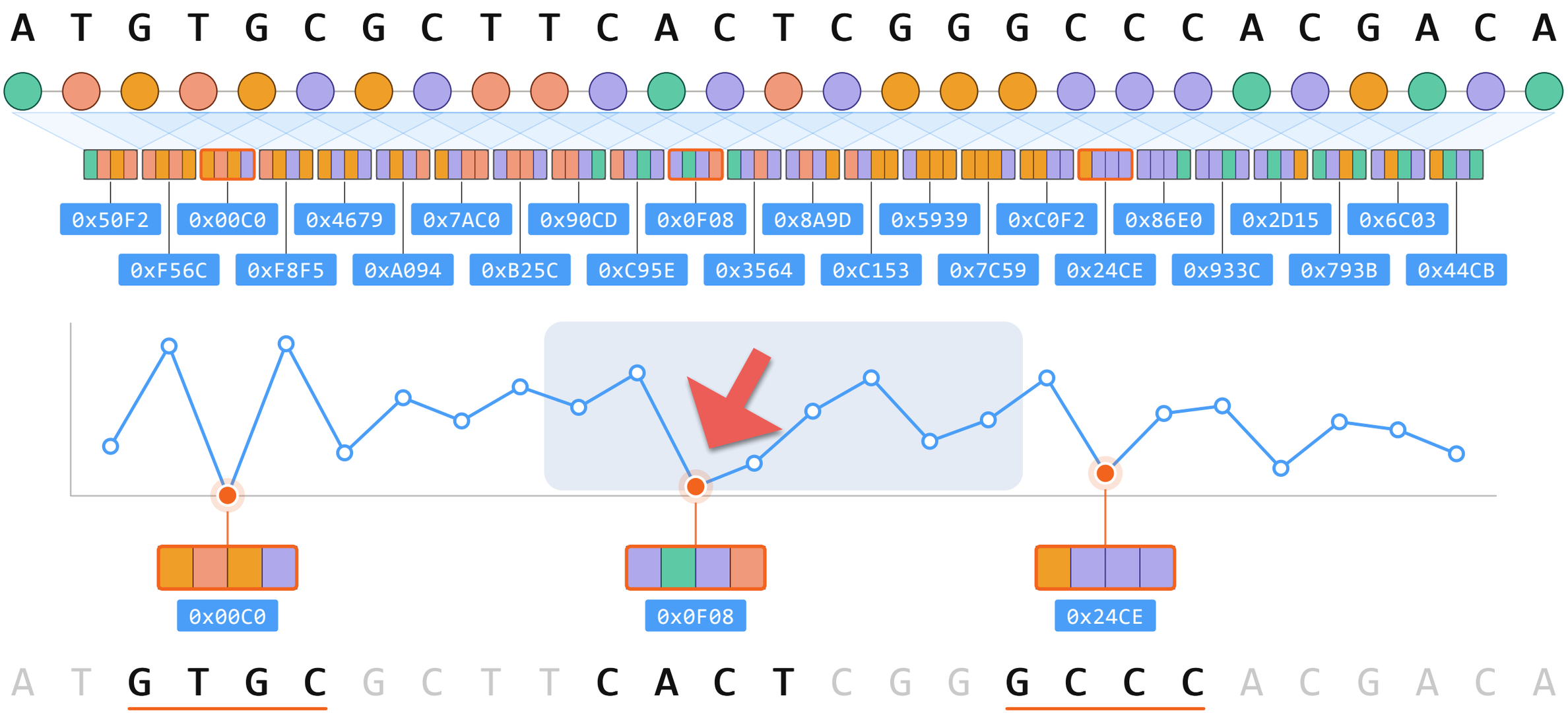


Image by Ben & Claude Sonnet 5.5

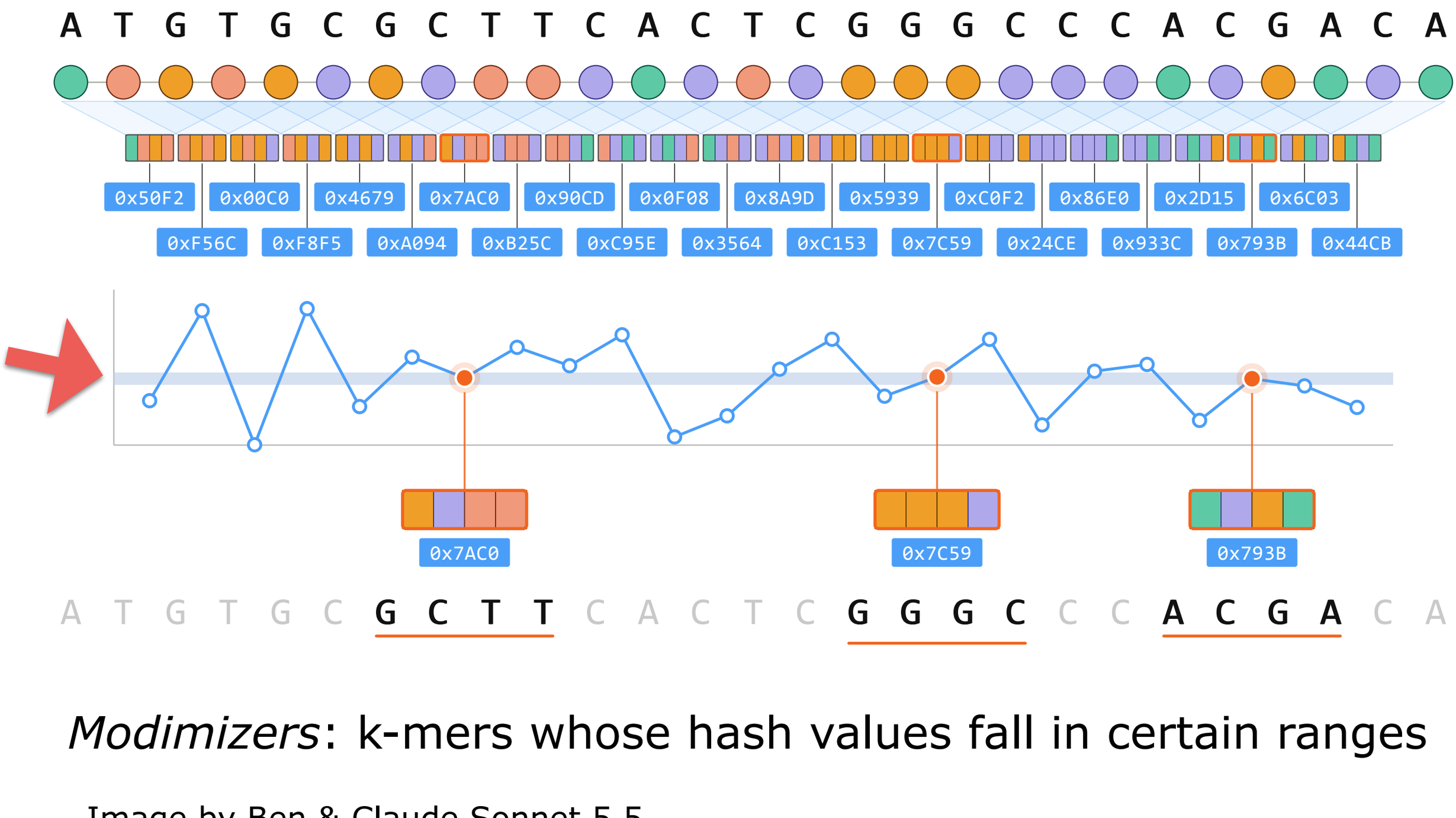
Another kind of “sketching”… or “digesting”



Minimizers: k-mers that are minimal in their neighborhoods

Image by Ben & Claude Sonnet 5.5

Another kind of “sketching”… or “digesting”



Modimizers: k-mers whose hash values fall in certain ranges

Image by Ben & Claude Sonnet 5.5

Sketching

Fixed, small space budget

Output is a “sketch” that packs the estimators in as tight as possible

Sketches tell us about cardinalities and coincidences…

…which in turn tell us about sequence similarity

Mash, Dashing, Bindash, …

Digesting

Space budget grows with length of sequences

Output is a “digest” built from k-mers sampled throughout

Digests shrink the representation and change the “alphabet”

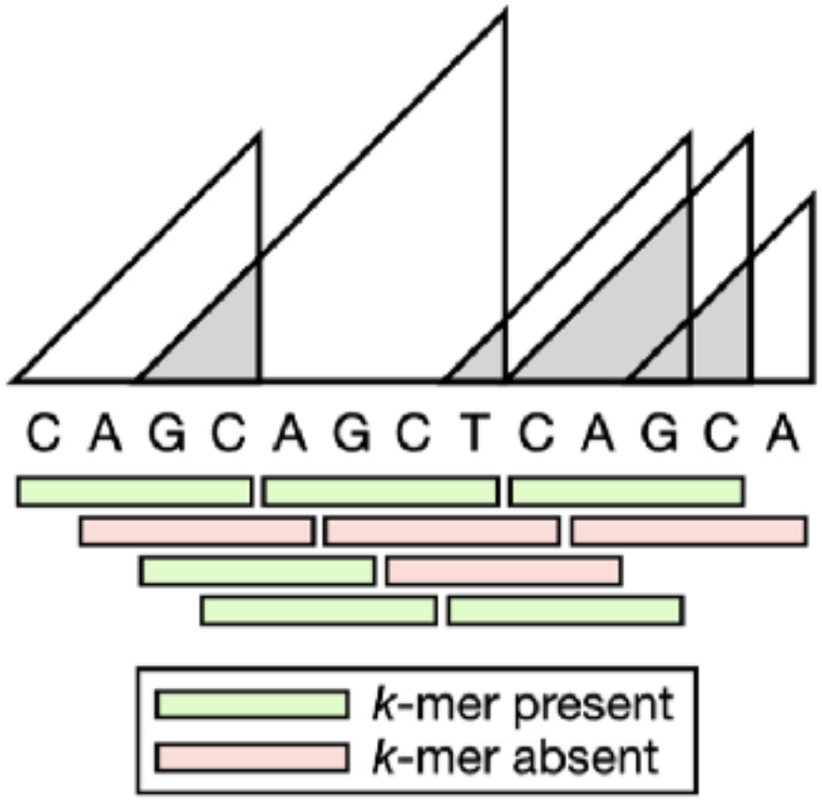
Allow for more efficient string comparisons

FracMinHash, sourmash, minimap, FastANI, ModDotPlot, …

If not k-mers/sketching, then what?

Full text indexes

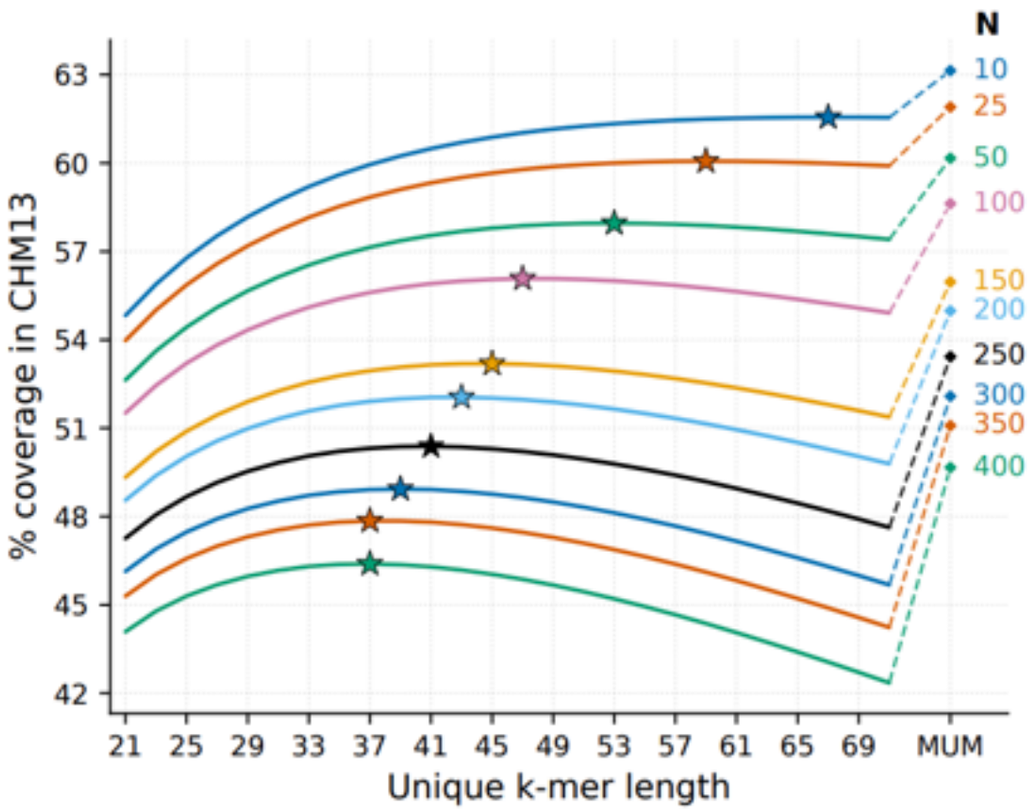
Flexible-length exact matches imply k-mers memberships for **all** k-mer lengths at once, sacrificing no contiguity…



…but are more costly (for now) 🤔 🤔 🤔

Hwang, S., Brown, N. K., Ahmed, O. Y., **Jenike, K**. M., Kovaka, S., Schatz, M. C., & Langmead, B. (2025). MEM-based pangenome indexing for k-mer queries. Algorithms for Molecular Biology, 20(1), 3.

If not k-mers/sketching, then what?



MUMs for conservation

We may need k-mers that "chop up" the pangenome in useful ways. MUMs subsume all possible choices of k-mers that are good at this.

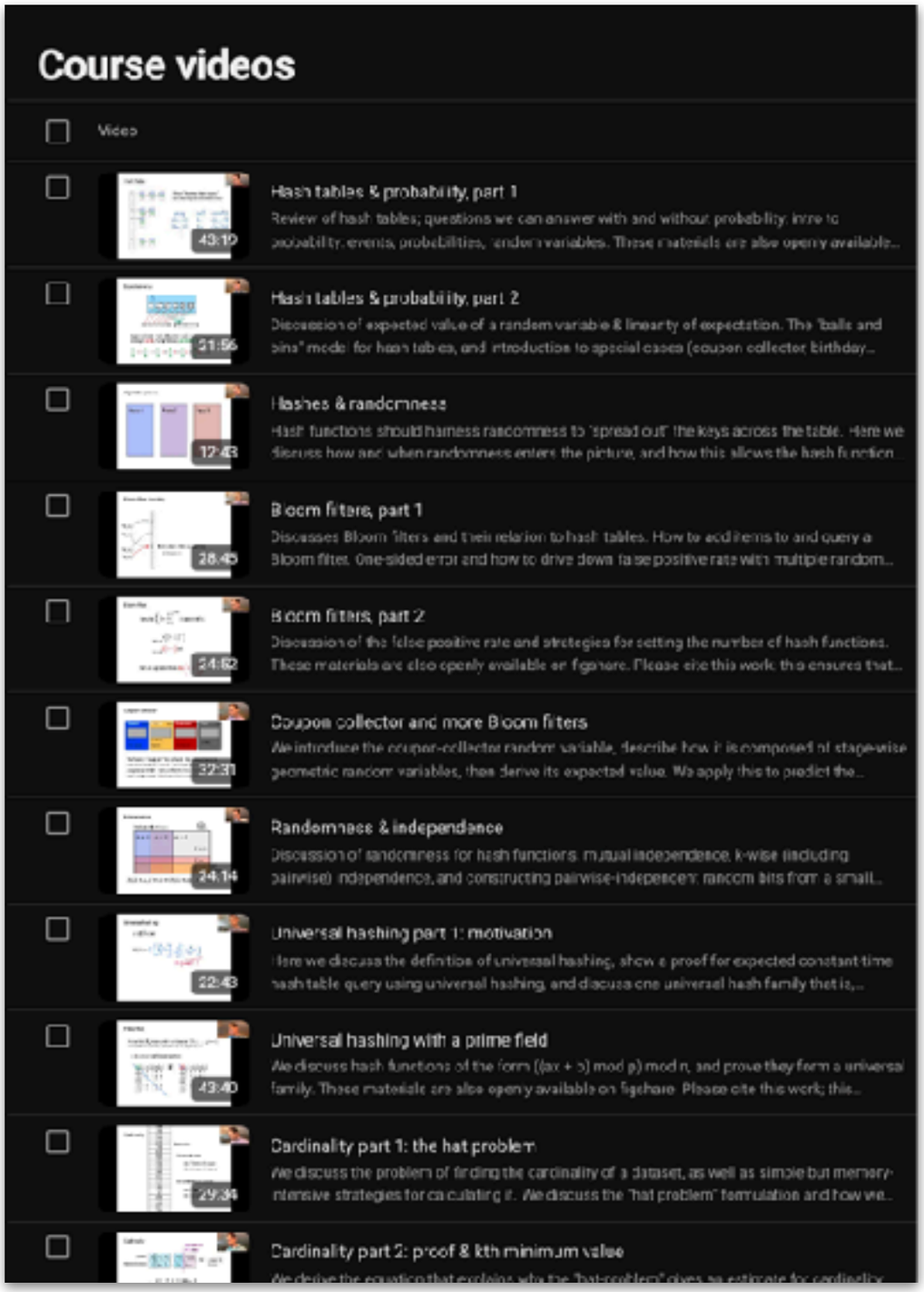
Shivakumar, V. S., Langmead, B., & Human Pangenome Reference Consortium. (2026). Navigating the pangenome coordinate system with Shredtools. bioRxiv, 2026-07.

Shivakumar, V. S., & Langmead, B. (2025). Mumemto: efficient maximal matching across pangenomes. Genome Biology, 26(1), 169.

Other resources

See my YouTube “course” on Sketching and randomized data structures

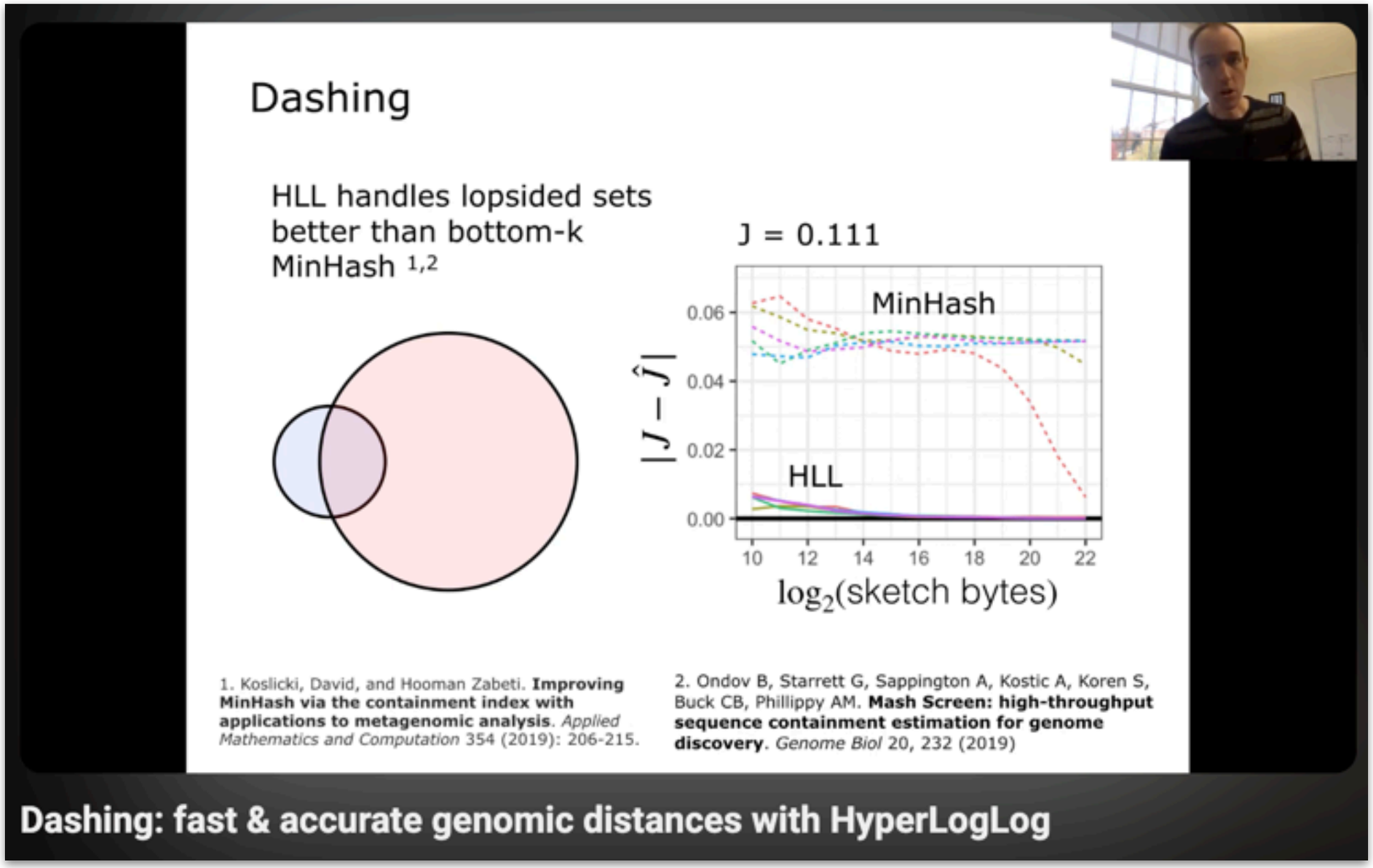
Covers hash tables, Bloom filters, cardinality, MinHash, CountMin



https://www.youtube.com/playlist?list=PL2mpR0RYFQsAR5RyB54FyEE9vUiGtCSZM

Or just go to BenLangmead channel and look for courses

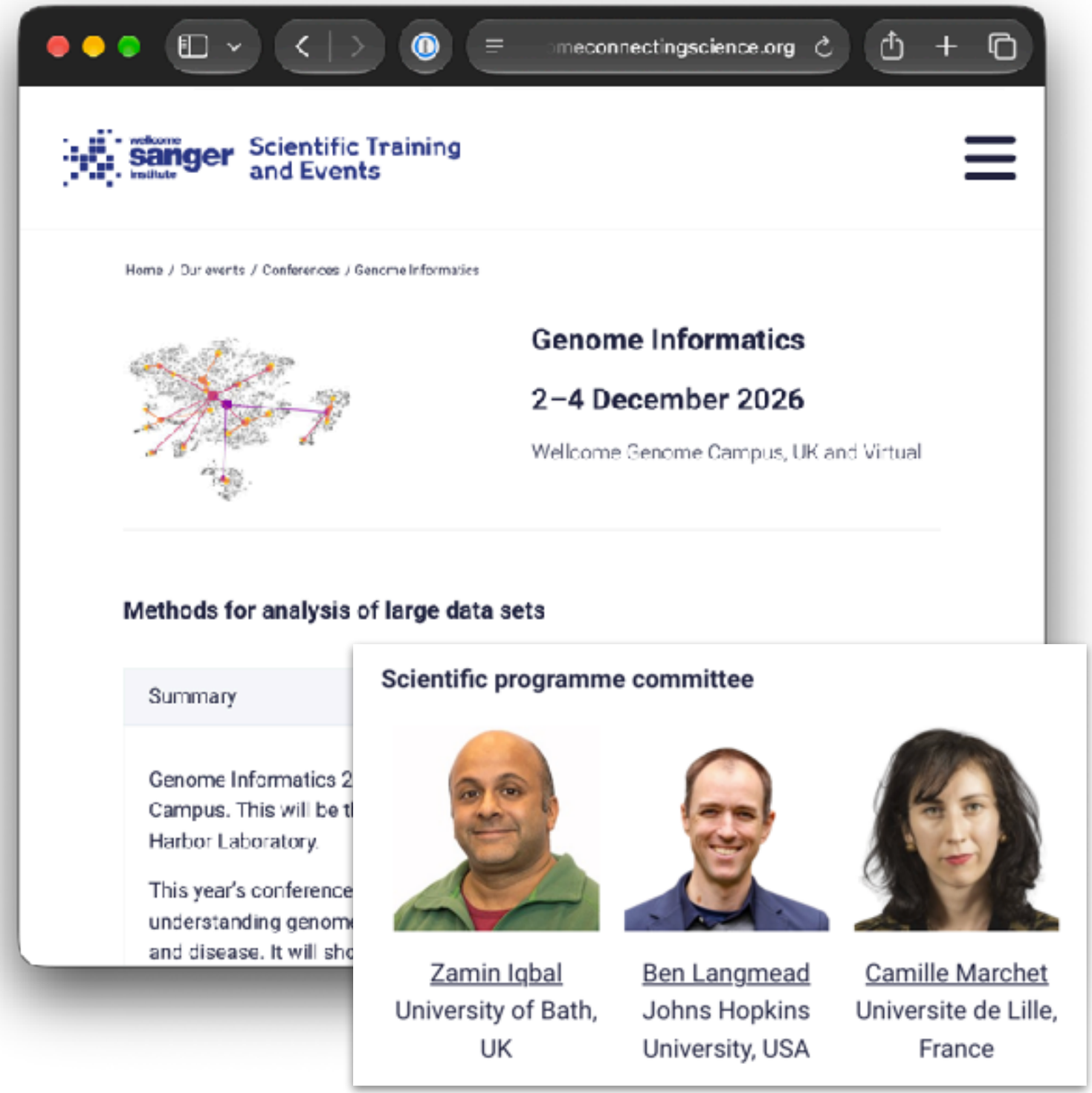
Talk on HLL & Dashing



https://youtu.be/1y9wqQUTvC4?si=qZok1YMUK8EfzV4R

Or just go to BenLangmead channel and search for "Dashing"

Thank you / questions



I described work from many people; my work on sketching is funded by NIH grants **R35GM139602** & R01HG011392

https://coursesandconferences.wellcomeconnectingscience.org/event/genome-informatics-20261202/